CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 02



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic Instruction

Topic ISA

Topic Types of Instruction

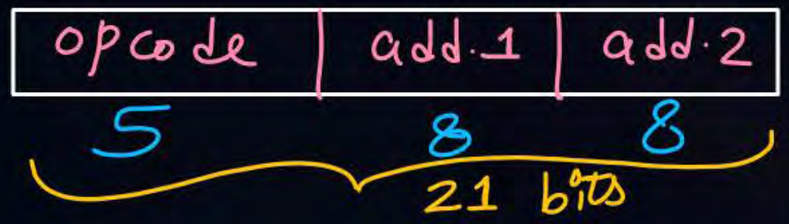
Topics to be Covered



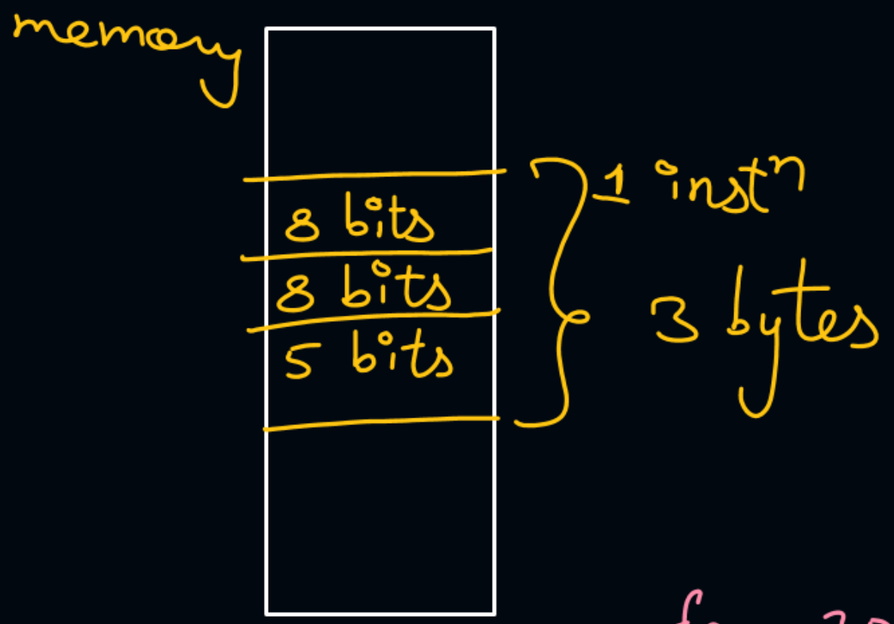
Topic	Instruction

[NAT]

#Q. Consider a digital computer which supports 32 2-address instructions. Consider the address length is 8-bits. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 300 instructions, then amount of memory required to store the program text is 900 bytes?



no. of instns supported = 32 $\Rightarrow$ opcode = $\lceil \log_2 32 \rceil$ = 5 bits



for 300 instns = 300 * 3B

= 900 B

[NAT]



#Q. A processor has 50 distinct instructions and 16 general purpose registers. Each instruction in system has one opcode field, 2 register operand fields and a 10 bits memory address field. The length of the instruction is 24 bits?

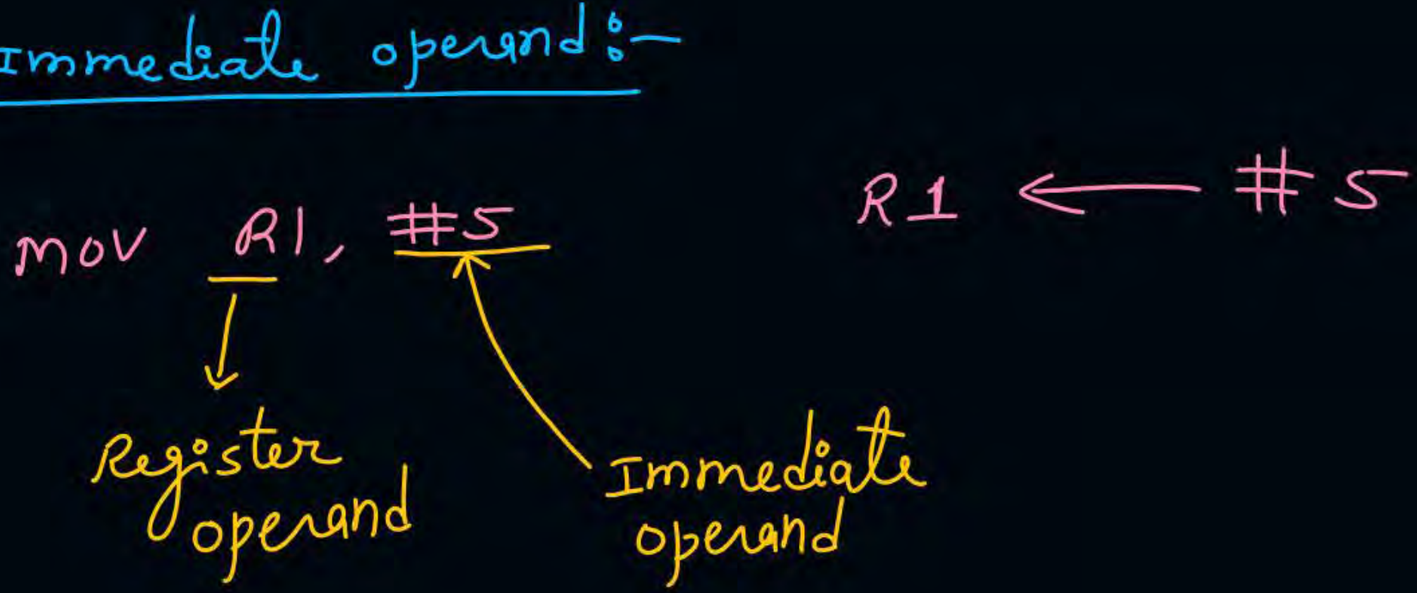
opcode	Reg1	Reg2	mem. Address
6	4	4	10

24 bits

No. of instructions supported = 50 $\Rightarrow$ opcode $= \lceil \log_2 50 \rceil = 6$ bits

No. of registers = 16 $\Rightarrow$ Register operand $= \lceil \log_2 16 \rceil = 4$ bits

Immediate operand :-

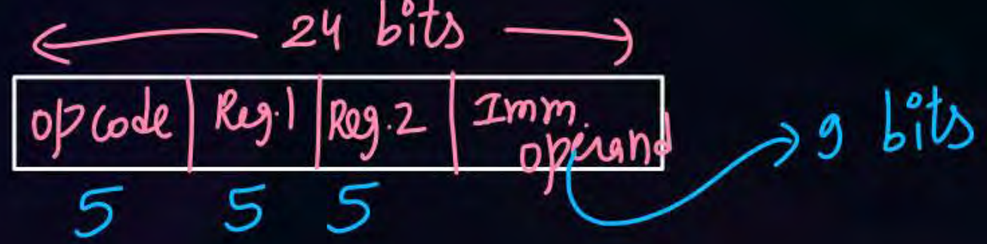


[NAT]

#Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24- bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9 ?

no. of instns = 20 $\Rightarrow$ opcode = $\lceil \log_2 20 \rceil$ = 5 bits

no. of reg. = 32 $\Rightarrow$ Reg. operand = $\lceil \log_2 32 \rceil$ = 5 bits



[NAT]

#Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24- bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9 ?

In above question: Assume that immediate operand field is an unsigned number, What is its maximum and minimum value possible?

max = 511

min = 0

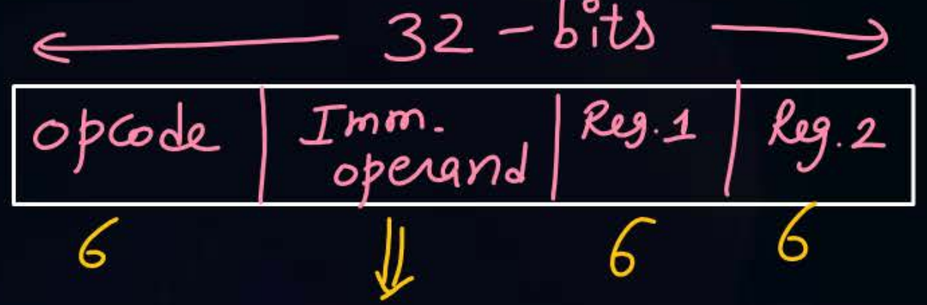
	max value	min value
for n bits unsigned value ⇒	$2^n - 1$	0
for 9 bits —11— ⇒	$2^9 - 1 = 511$	0

for n bits value	min	max
Unsigned	0	$2^n - 1$
Sign-magnitude	$-(2^{n-1} - 1)$	$+(2^{n-1} - 1)$
1^s complement	$-(2^{n-1} - 1)$	$+(2^{n-1} - 1)$
2^s complement	-2^{n-1}	$+(2^{n-1} - 1)$

[NAT] GATE-2014

ALU input size = 1 word = 32 bits

#Q. A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is 16383.



max value = $2^{14} - 1 = 16383$

No. of registers = 64 ⇒ Reg. operand = 6 bits

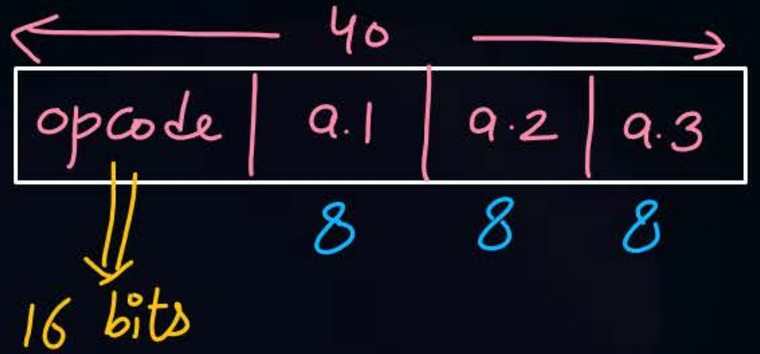


No. of instns = 45 ⇒ opcode = $\lceil \log_2 45 \rceil$ = 6 bits

[NAT]

byte addressable

#Q. Consider a system which support only 3 address instructions only, and supports 256B memory. If the instruction size is 40-bits then maximum & minimum number of instruction supported by the system are?



No. of cells in mem. $= \frac{256\,B}{1\,B}$

$= 256 = 2^8$

address = 8 bits

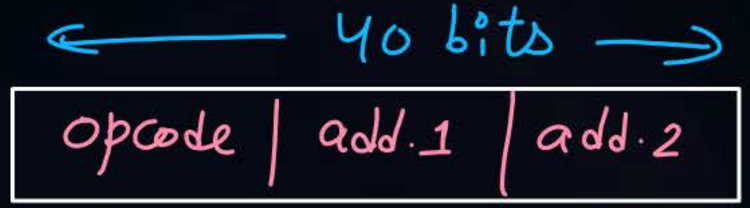
min = 1

max = $2^{16} = 65536$

Ans.

[NAT]

#Q. Consider a system which support only 2 address instructions only and supports byte addressable memory. If the instruction size is 40-bits and number of distinct instructions supported by the system is 256 then the maximum size of memory supported by the system is 64 Kbytes?



No. of instns supported = 256

opcode = $\lceil \log_2 256 \rceil$ = 8 bits

$8 + 2x = 40$

$x = 16 \Rightarrow$ max no. of cells in mem. = 2^{16}

max mem size = $2^{16} * 1B = 64KB$

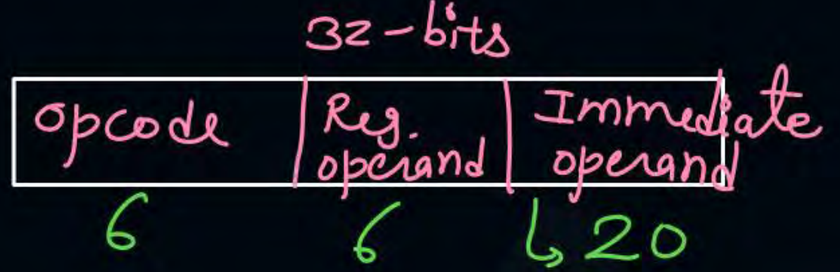
[NAT] GATE-2025

Reg. operand = 6 bits

opcode = 6 bits

#Q. A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction?

ADD R1, #25 // R1 = R1 + 25



A 16

B 20

C 22

D 24



2 mins Summary

Topic Instructions



Happy Learning

THANK - YOU